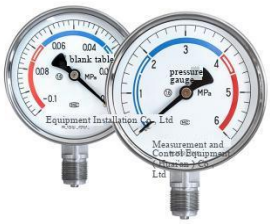
Guide for Standardized Calibration and Precise Detection and Measurement of Stainless Steel Pressure Gauges

To standardize the calibration, testing, and measurement of pressure gauges and ensure their precise measurement and reliable data, this operation specification is formulated. This calibration utilizes the ALKB9651 micro pressure calibration bench as the pressure source, paired with the ALKC400E digital pressure calibrator as the standard gauge, to rigorously conduct full-range calibration under both atmospheric pressure and vacuum conditions. The process encompasses preliminary environmental preparation, point-by-point pressure increase/decrease, light tapping for reading, and error calculation, aiming to mitigate the risk of pressure shock. Additionally, it emphasizes regular metrological verification and maintenance of equipment. By standardizing operations to identify instrument errors, we ensure stable measurement performance and provide reliable data support for industrial production pressure monitoring.

1. Calibration tool for pressure gauge

Name	Micro-pressure calibration bench	Digital pressure calibrator	Stainless steel pressure gauge
Model number	ALKB9651	ALKC400E	Y-100BF
Technical parameters	Suppression method: manual Pressure range: -90~600kPa Medium: Clean air Output ports: 2 Connection: M20×1.5 internal Net weight: 2.3kg Product size: 230×240×200mm	Pressure: -100~600kPa Accuracy: 0.02%FS Temperature compensation: with temperature compensation Current measurement: ±30.0000mA Voltage measurement: ±30.0000V DC output: 24VDC Communication interface: RS-232/RS-485 Power supply: built-in rechargeable battery Connection: M20×1.5 external Net weight: 0.6kg Overall dimensions: 110×50×185mm	Pressure: -100~0kPa/0~600kPa Accuracy: Level 1.6 Material: stainless steel Dial diameter: φ100mm Connection: M20×1.5 external Net weight: 0.35kg Product size: 100×42×140mm
Product appearance			

The entire verification process must strictly adhere to the current metrological technical specifications. The main basis includes JJG52-2013 "General Pressure Gauges, Pressure Vacuum Gauges, and Vacuum Gauges with Elastic Elements", which is applicable to the initial verification, subsequent verification, and in-service inspection of general pressure gauges, pressure vacuum gauges,

and vacuum gauges with elastic elements, covering locally indicating pressure measuring instruments commonly used in industry.

II. Detailed Explanation of Operational Procedures

(I) Preliminary preparations

1. Environmental requirements

The temperature should be controlled at (20±2)℃, with a relative humidity of ≤85%RH, and there should be no condensation. Ensure that the site is free from vibration, dust, and corrosive gases.

2. Tool connection

• Connect the standard gauge ALKC400E and the verified pressure gauge Y-100BF in a sealed manner to the ALKB9651 output port.

(II) Pressure gauge calibration process

1. Voltage boosting process

• First, rotate the pressure adjusting handle counterclockwise until it cannot be adjusted any further. Then, rotate the fine adjustment knob counterclockwise until it cannot be adjusted any further. Finally, close the pressure relief valve.

• Rotate the pressure adjusting handle clockwise to apply pressure until it reaches around the verification value, then use fine adjustment to slowly adjust it to the verification point of 150kPa. After the pressure stabilizes and the standard indicated value no longer fluctuates, check the pressure value of the verified pressure gauge. Then, gently tap the side of the verified pressure gauge's casing with your finger (tapping on the glass or hitting hard is strictly prohibited), observe the pointer position, and record the indicated value after the tap stabilizes. Use the indicated value after the tap as the actual measured verification reading for this pressure point, and calculate the error.

• Perform the same operation at verification points of 300kPa, 450kPa, and 600kPa, check the pressure values of the verified pressure gauge, tap the verified pressure gauge, and record the indicated value after the tap stabilizes.

2. Decompression process

• Rotate the pressure regulating handle counterclockwise to reduce the pressure until it reaches around the verification value, then use fine adjustment to gradually adjust it to the verification point of 450kPa. Once the pressure stabilizes, tap the verified pressure gauge and record the value.

• Perform the same operation with the verification point adjusted to 300kPa and 150kPa, check the pressure value of the verified pressure gauge, tap the verified pressure gauge, and record the indicated value after the tap becomes stable.

3. Pressure relief process

Open the pressure relief valve, gradually release the pressure inside the pipeline, and when the pressure relief reaches 0, the measurement is complete.

(III) Vacuum calibration process of pressure gauge

1. Voltage boosting process

First, remove the verified meter for measuring positive pressure, install the verified meter for measuring negative pressure, turn the pressure regulating handle clockwise until it stops, and then close the pressure relief valve.

• Use the pressure regulating handle to slowly adjust counterclockwise until it reaches around the verification value, and then use fine adjustment to gradually adjust to the verification point of -25kPa. Once the pressure stabilizes, tap the verified pressure gauge and record the value.

• Perform the same operation with the verification points adjusted to -50kPa, -75kPa, and -90kPa, check the pressure values of the verified pressure gauge, tap the verified pressure gauge, and record the indicated value after the tap becomes stable.

2. Decompression process

• Rotate the pressure regulating handle clockwise to reduce the pressure until it reaches around the verification value, then use fine adjustment to gradually adjust it to the verification point of -75kPa. Once the pressure stabilizes, tap the verified pressure gauge and record the value.

• Perform the same operation and adjust the verification point to -50kPa and -25kPa, check the pressure value of the verified pressure gauge, tap the verified pressure gauge, and record the indicated value after the tap becomes stable.

3. Pressure relief process

Open the pressure relief valve, gradually release the pressure inside the pipeline, and when the pressure relief reaches 0, the measurement is complete.

III. Precautions

Standardized use

• During verification, it is important to ensure that the tolerance of the standard instrument is less than that of the instrument being verified, to ensure accurate and reliable verification results that meet the metrological specifications.

• Operations should follow standard procedures. The ALKB9651 micro-pressure calibration platform should be pressurized and depressurized slowly to avoid pressure shock.

• Increase the pressure to the verification point and record the indicated value after the standard gauge stabilizes. For the verified pressure gauge, gently tap the gauge case after the pressure stabilizes, and take the indicated value after tapping as the accurate reading to ensure accurate recording.

• After the inspection is completed, slowly release the pressure to zero and ensure there is no residual pressure before removing the inspected instrument.

Maintenance and calibration cycle

• Regularly conduct metrological verification on the digital pressure calibrator ALKC400E (it is recommended to do so once a year) to ensure that the measurement accuracy meets the standards.

• When the pressure gauge is not in use, the power should be turned off. If it is not used for a long time, it should be charged once every three months.

• If the micro-pressure stage is not used for a long time, it should be depressurized and stored, covered with a cloth cover to prevent dust from entering the instrument.

IV. Conclusion

This pressure gauge calibration strictly follows the standardized process, including pre-preparation such as temperature and humidity control, as well as step-by-step pressure increase/decrease under both atmospheric pressure and vacuum conditions, light tapping for reading, and error calculation. During the operation, the risk of pressure shock is avoided, and emphasis is placed on regular metrological verification and scientific maintenance of equipment. This effectively identifies instrument indication errors, ensures measurement accuracy, and provides support for pressure measurement in industrial production and scientific research experiments.